



Module A – Cybersecurity Fundamentals

A1 – Why Cybersecurity Exists

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Acknowledgments

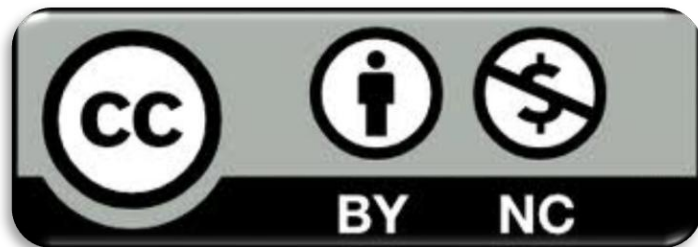
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- This material is used and adapted within the course “Cybersecurity and National Defense”.



Learning goals

After this lecture, you should be able to:

- Explain why cybersecurity became a necessity
- Describe how connectivity reshaped risk
- Understand cybersecurity as a ***systemic issue***
- Recognize why cyber incidents can scale to ***national relevance***



One click can unlock convenience... or unleash chaos

Convenience

Digital services scale
convenience, speed, and reach:

- Online banking 
- Online shopping 
- Chat applications 
- Healthcare portals 
- Smart devices 
- ...

Chaos

The same infrastructure also
scales *failures, attacks, and
disruption:*

- fraud
- ransomware
- outages
- leaked data
- deep fake
- ...

From isolated systems to pervasive cyberspace

1945



The huge ENIAC (**E**lectronic **N**umerical **I**ntegrator and **C**omputer) aka "the Giant Brain" (**1945**) filled a room and required six programmers.

Isolated Computing (1940s–1960s)

- Early computers were **isolated, centralized machines**
- Physical access was required to interact with the system
- Users were few, trusted, and co-located
- Failures and misuses were **local and contained**



Security assumptions were physical, not digital

From isolated systems to pervasive cyberspace

1948

1945



The Manchester Baby (1948), the first computer to store programs digitally.

The Stored-Program Concept

- Programs and data are stored in the **same memory**
- Instructions become **modifiable digital objects**
- This abstraction enables:
 - software flexibility, reusable logic
 - reprogrammable machines, innovation
 - software exploitation





From isolated systems to pervasive cyberspace

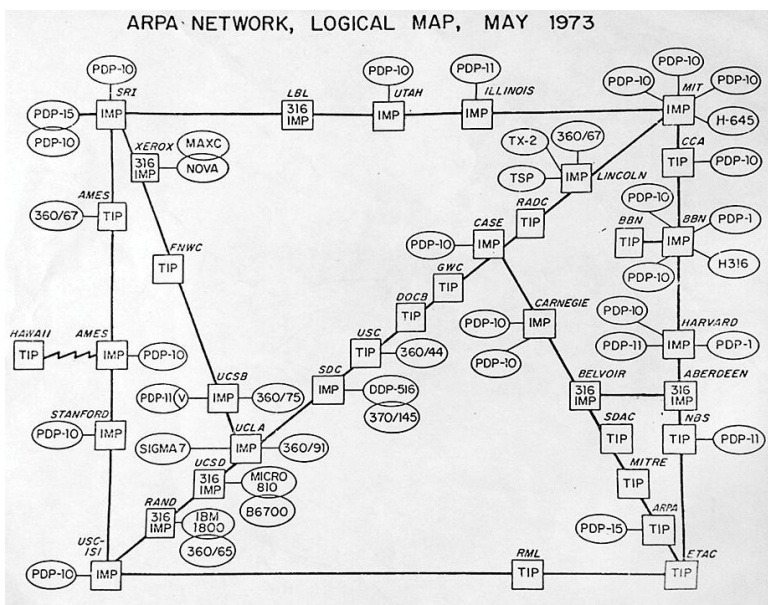


1945

1948

1969

ARPA NETWORK, LOGICAL MAP, MAY 1973



ARPANET map (1973)

Networking Changes the Game

- Networking enables **remote collaboration**
- ARPANET introduced packet switching
- Trust assumptions expand beyond a single room
- Distance stops being protection

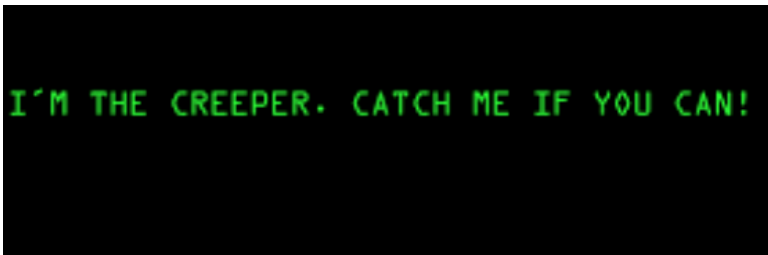


From isolated systems to pervasive cyberspace



Creeper: the first self-propagating program

- Developed by Bob Thomas at BBN
- Ran on ARPANET
- Self-propagating program
- **Not malicious**, but demonstrated a new capability
- Showed that code could move autonomously across a network



Screenshot of Creeper message.



Bob Thomas



Code mobility is a security problem

From isolated systems to pervasive cyberspace



Reaper: the first defensive response

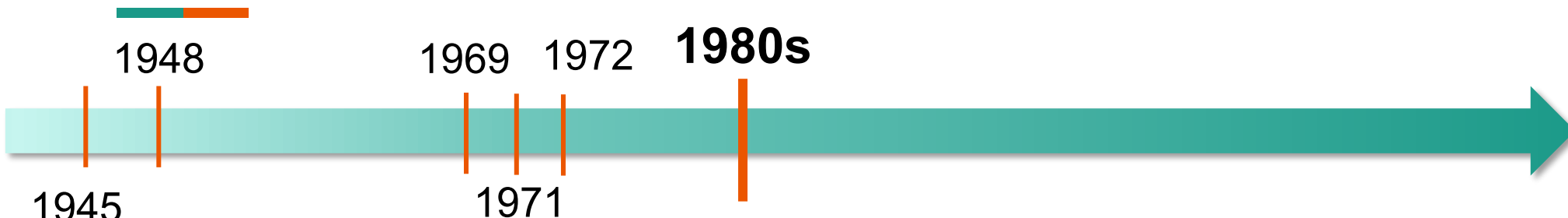
- Developed by Ray Tomlinson to locate and remove Creeper
- First example of **active defense**
- Introduced the idea of “defensive software”
- Attack and defense start to **co-evolve**

Ray Tomlinson (inventor of email)

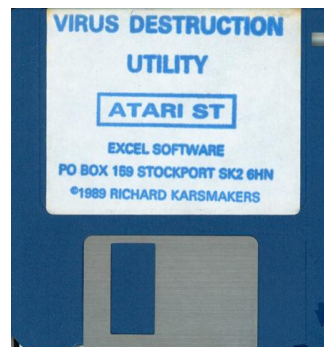
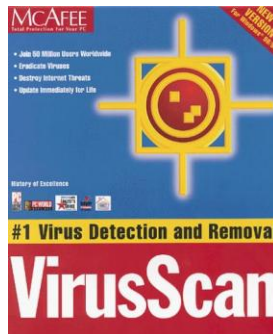


Defense exists because threats scale

From isolated systems to pervasive cyberspace



Jhon David McAfee,
developer of one of
the first antivirus
programs for home
users



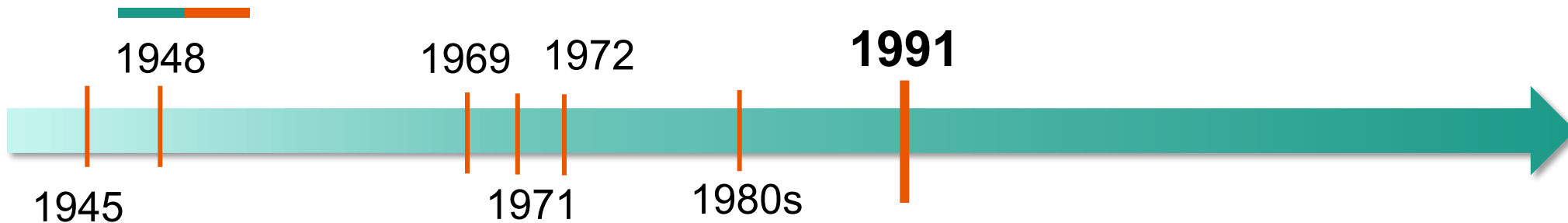
The 1980s: the birth of commercial defense

- Personal computers become widespread
- Viruses spread via floppy disks
- First commercial antivirus products appear
- Security becomes a **market**



Security follows adoption

From isolated systems to pervasive cyberspace



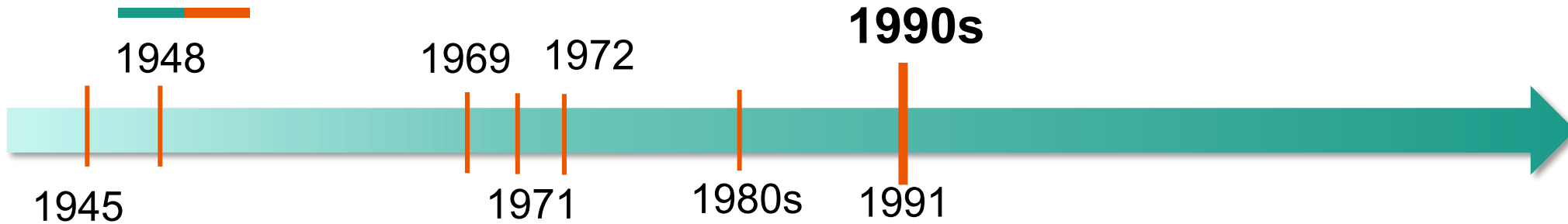
The Birth of the World Wide Web

- Web makes the Internet **usable by everyone**
- Browsers, URLs, HTTP lower the entry barrier
- Mass adoption begins
- Digital identity starts to matter

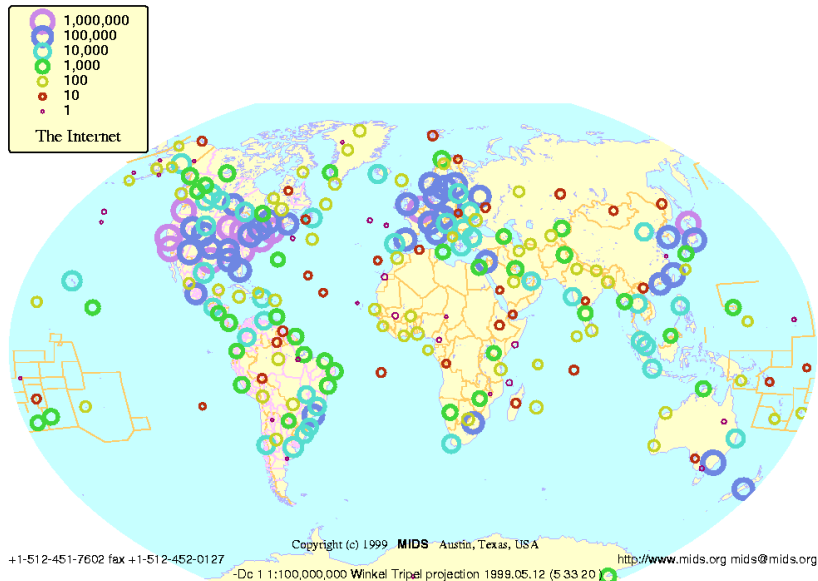
Tim Berners-Lee, the inventor of the World Wide Web, HTML, the URL system, and HTTP



From isolated systems to pervasive cyberspace



The Internet Jan 1999



The 1990s: the world goes online

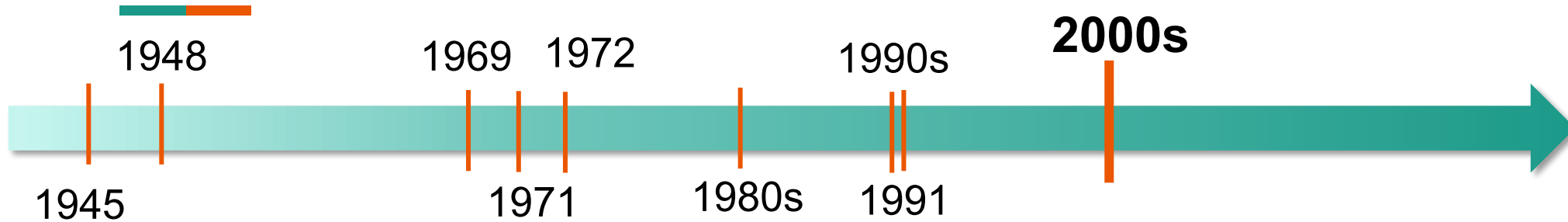
- Internet becomes public and commercial
- Internet supports: commerce, communication, information exchange
- Personal data moves online
- Identity theft and cybercrime emerge
- Attacks become economically motivated
- Firewalls and antivirus become standard



When value moves online, crime follows



From isolated systems to pervasive cyberspace



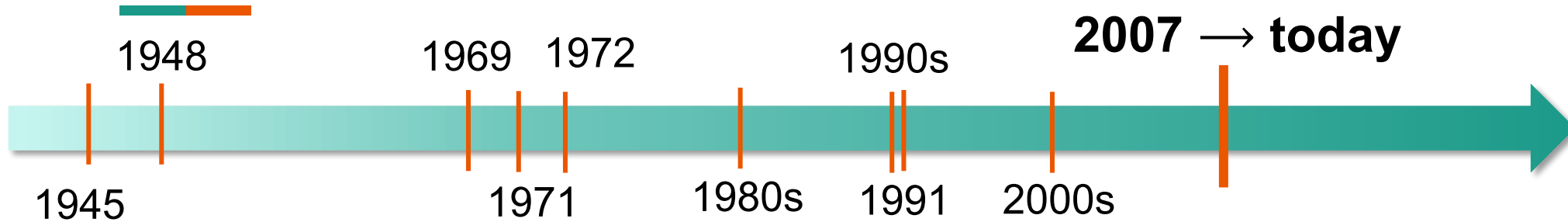
The 2000s: threats diversify and professionalize

- Organized cybercrime
- Crime-as-a-Service
- Governments start reacting
- Security becomes strategic



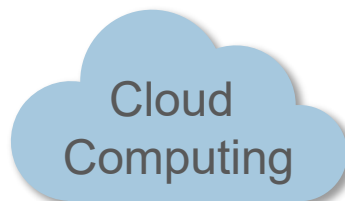
Cybersecurity becomes a national concern

From isolated systems to pervasive cyberspace



Pervasive Connectivity

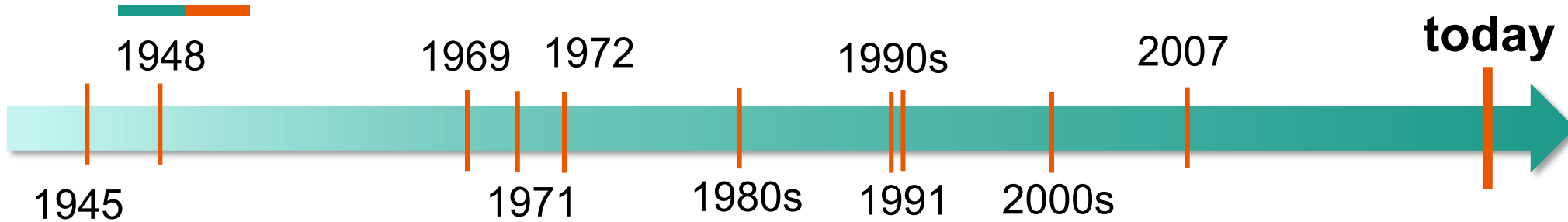
- Internet becomes:
 - mobile
 - always-on
 - embedded in physical systems
- Attack surface explodes



iPhone 1st generation, released
in the United States on June
29, 2007

Cybersecurity becomes a **systemic** problem

From isolated systems to pervasive cyberspace

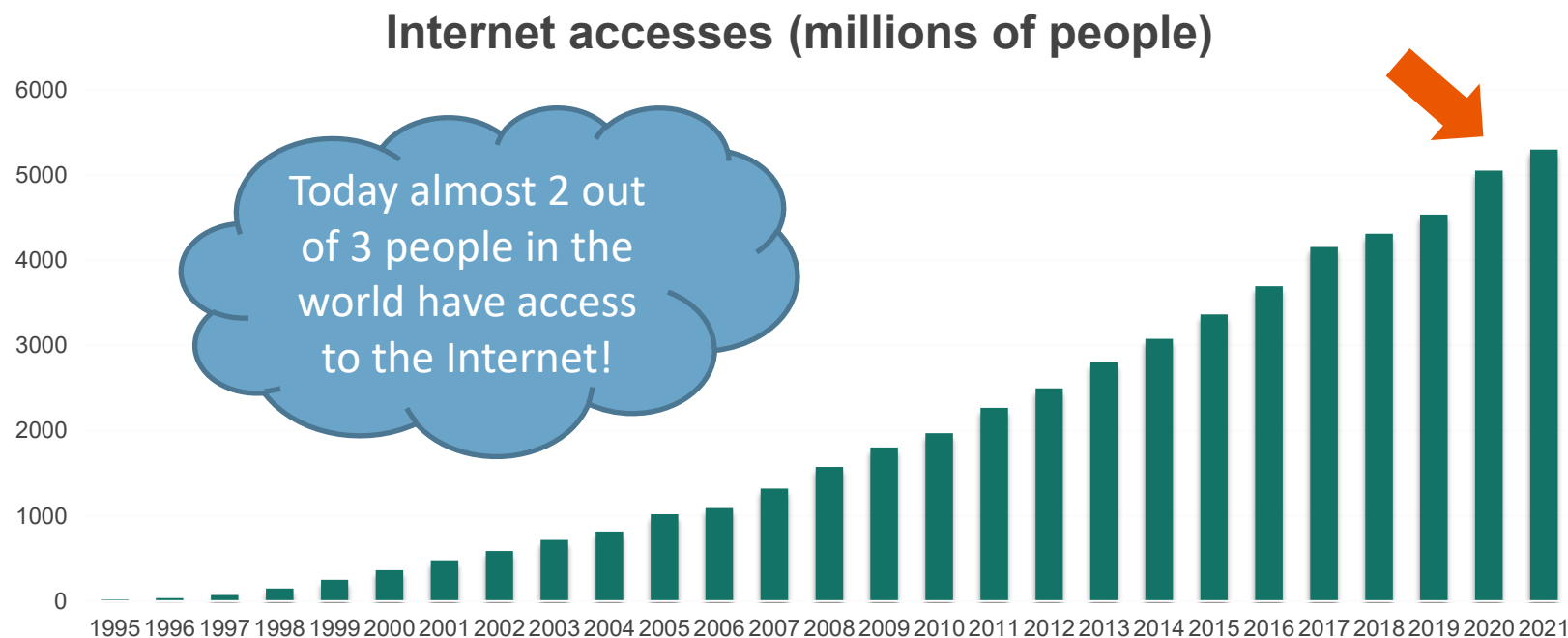


Cyberspace

- Cyberspace is a **socio-technical environment**
- Interdependencies dominate
- Failures propagate



How much is the Internet growing in the world?



[Nunzia Ciardi,
Vice Direttore Generale CAN - Agenzia per la Cybersicurezza Nazionale]

The use of technology in Italy



In 2009 in relation to the population:

only 15% were using smartphones

In 2022 in relation to the population:

129,7% has a mobile device connected

84,3% uses internet

71,6% uses social media

[Nunzia Ciardi,
Vice Direttore Generale ACN - Agenzia per la Cybersicurezza Nazionale]

What do we mean by “system”?



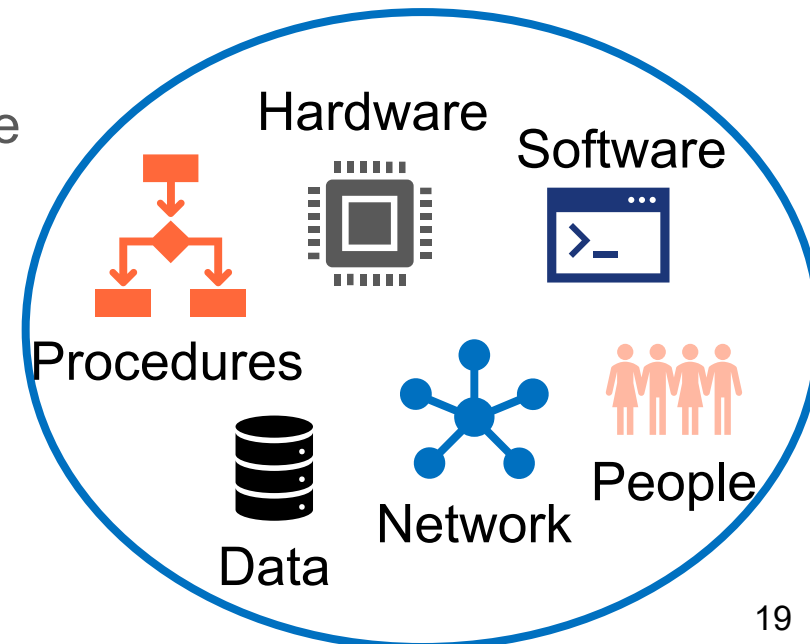
Security is no longer local, it is **systemic**, but ... what do we mean by “**system**”?

A system is not just a single device or application. A **system** is a set of interacting components that:

- perform a function
- exchange information
- and depend on each other to deliver a service

In cybersecurity, **the system includes:**

- hardware and software
- networks and data
- users and operators
- procedures and dependencies

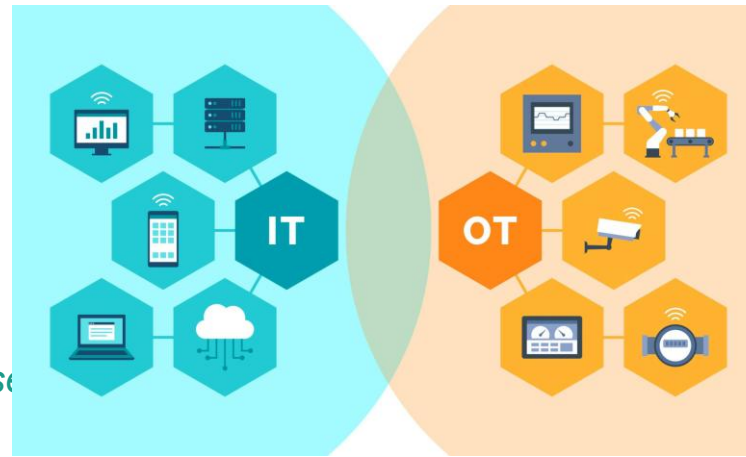


IT systems and OT systems



- **IT (Information Technology)** systems manage data, computation, and communication (e.g., servers, databases, cloud services)
- **OT (Operational Technology)** systems control physical processes (e.g., industrial controllers, sensors, actuators)

Modern infrastructures **combine IT and OT**, creating new dependencies, new attack paths, real-world physical consequences





Cybersecurity

- The computerization of our society and the digitization of goods and services, in both the public and private sectors, force us to pay a greater and greater attention to the security of both IT and OT assets
- Several definitions can be found in the literature
- Here we are going to focus just on three of them, highlighting different aspects:

user → system → nation



Cybersecurity -- definition 1 (operational)



Example of an operational, user-centric definition (UK NCSC):

- Cybersecurity is how individuals and organisations reduce the risk of cyber attack.
- Cybersecurity's core function is to protect the **devices** we all use (smartphones, laptops, tablets and computers), and the **services** we access - both online and at work - from theft or damage.
- It's also about preventing unauthorised access to the vast amounts of **personal information** we store on these devices, and online.

[<https://www.ncsc.gov.uk/section/about-ncsc/what-is-cyber-security>]



Cybersecurity -- Definition 2 (system-level)



Set of activities necessary to:

- protect networks, information systems, computer services, and electronic communications from cyber threats
- ensure their *confidentiality*, *integrity*, and *availability* (**CIA**), and also ensure their *resilience*.



Cybersecurity -- Definition 3 (national-scale)



Practice that allows an entity (organization, citizen, **nation**, ...) to protect:

- its **physical assets**
- *confidentiality, integrity and availability* of its information from threats that come from **cyberspace**.

[standard ISO/IEC 27000:2014 & ISO/IEC 27032:2012]

This definition connects cybersecurity to national-scale assets and cyber-physical impact.



Common core across definitions

- Despite different scopes and formulations, cybersecurity definitions converge on a shared core:
 - *confidentiality*, *integrity*, and *availability* (CIA)
 - *resilience*
 - *risk reduction*
- These concepts will be revisited and formalized in the next lecture

Cyberspace



Cyberspace is the complex environment resulting from the interaction of people, software, services, and networks connected to the Internet.

It is:

- dynamic and evolving
- shaped by human behavior and automation
- governed by technical, legal, and social rules

[standard ISO/IEC 27000:2014 e ISO/IEC 27032:2012]

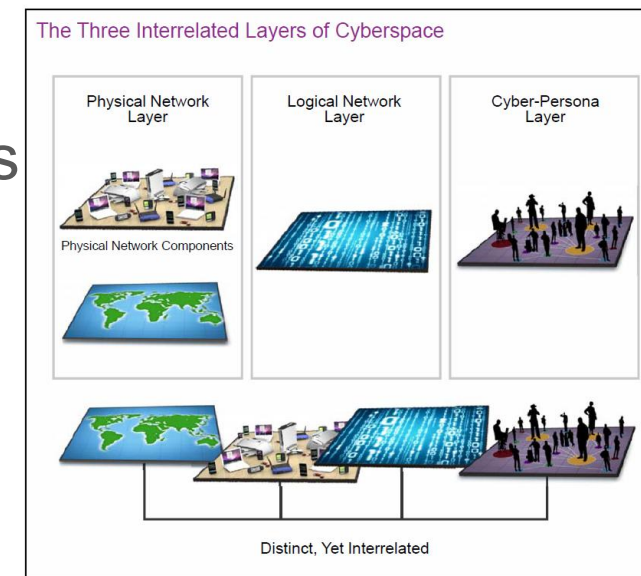


Figure I-1. The Three Interrelated Layers of Cyberspace

Cyberspace

A dynamic
socio-technical
environment

The Three Interrelated Layers of Cyberspace

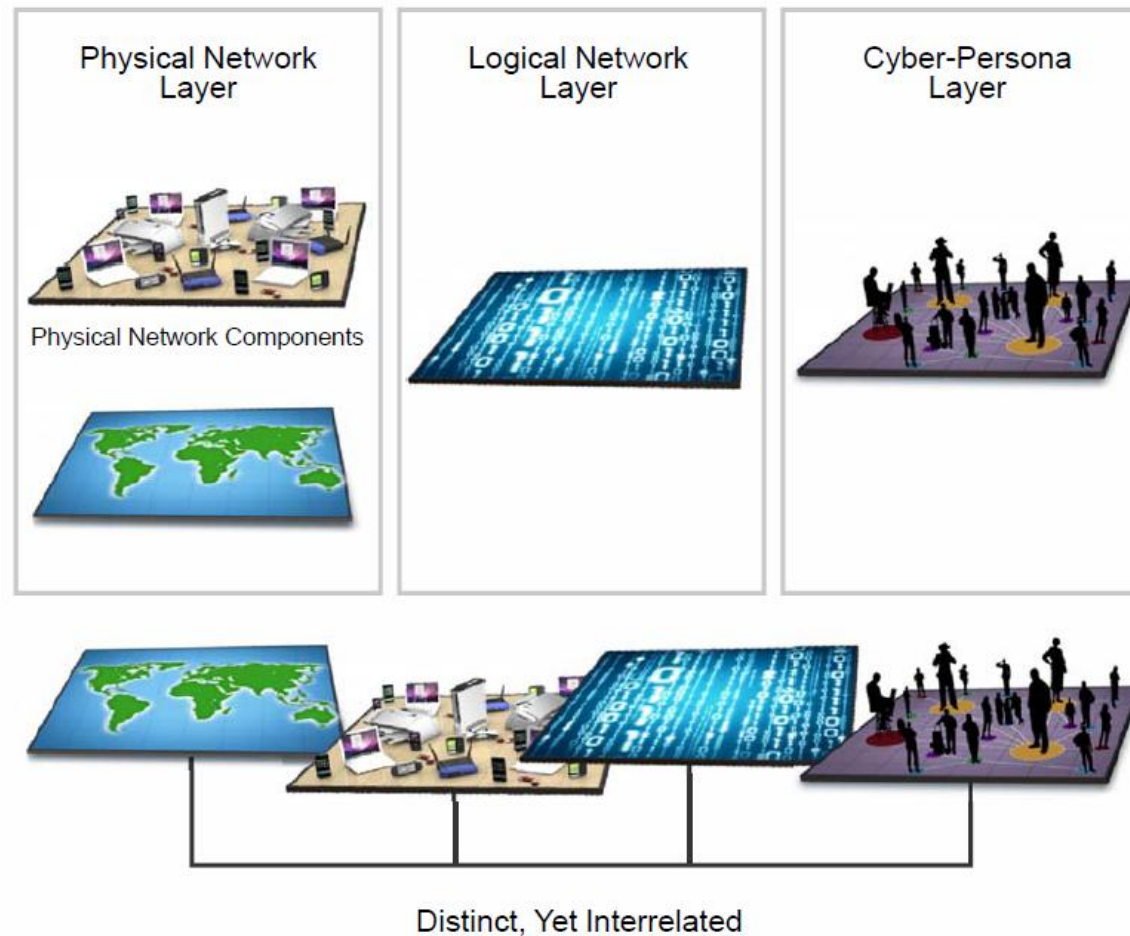


Figure I-1. The Three Interrelated Layers of Cyberspace



National Cyberspace



- The segment of the global network that falls within the nation's legal and administrative purview.
- Interlinked with the cyberspace, is distinct in its governance, under the jurisdiction of national laws and regulations, and is shaped by the country's unique economic, cultural, and political context.

[R.Baldoni - CHARTING DIGITAL SOVEREIGNTY: A SURVIVAL PLAYBOOK]



National Cyberspace

- Can be seen as a portion of the cyberspace, defined by the digital infrastructures within a nation's geographical boundaries and by its interconnected citizen, businesses and government.
- It includes all devices operating within the nation; the software running on these devices; the data generated, stored, or transiting through this infrastructure.
- National cyberspace is where security becomes a matter of *sovereignty, law, and coordinated response*.



Cyberspace – Complexity

The *most complex* thing that man has ever built:

- networks of networks
- software/protocol stratification
- heterogeneity of devices and terminals
- ...

Security *consequences*:

- Complexity creates hidden dependencies
- Small changes can have systemic effects
- Security failures often emerge gradually, until they become visible

Human interaction

Cloud / Services

Applications

Middleware

Operating System

Firmware

Hardware



Software – Complexity

- The Linux kernel exceeds 30 million Lines of Code (LoC)
 - Linux distributions exceed 100 million LoC
 - Microsoft Windows systems range between 30 and 45 million LoC
 - Large enterprise platforms contain hundreds millions LoC
 - Entire computing ecosystems can reach billions LoC
 - Comprehensive formal verification is infeasible
- ⇒ **Systems are vulnerable**
- More than 2,800 security vulnerabilities have been discovered in the Linux kernel alone
 - approximately 900 are of “high” or “critical” severity

Cyberspace – Pervasiveness

- Cyberspace is no longer confined to computers and networks: it pervades physical environments, daily life, and critical services





Why Cybersecurity Became a Necessity

- Digitalization made **connectivity** the default condition
- Connectivity transformed systems into **interdependent ecosystems**
- Interdependence amplified both failures and attacks
- Security moved from a technical issue to a **systemic necessity**

✦ Cybersecurity is a response to systemic digital risk



Some “*symptomatic*” examples

- The following examples illustrate how cybersecurity issues manifest in everyday and critical contexts



PRIVACY AND SECURITY FANATIC

By [Ms. Smith](#), Network World | FEB 12, 2017 8:15 AM PT

About

Ms. Smith (not her real name) is a freelance writer and programmer with a special and somewhat personal interest in IT privacy and security issues.

University attacked by its own vending machines, smart light bulbs & 5,000 IoT devices

A university, attacked by its own malware-laced soda machines and other botnet-controlled IoT devices, was locked out of 5,000 systems.



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THE INTERNET OF HACKABLE THINGS

Internet of Things Teddy Bear Leaked 2 Million Parent and Kids Message Recordings



LORENZO FRANCESCHI-BICCHIERAI

Feb 27 2017, 10:00pm

A company that sells “smart” teddy bears leaked 800,000 user account credentials—and then hackers locked it and held it for ransom.

UPDATE, Feb. 28, 12:25 p.m. ET: After this story was published, a security researcher revealed that the stuffed animals themselves could easily be hacked



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Attack on surveillance cameras a warning over security, ethics

The attack on a video surveillance startup by a hacktivist group raises questions not just over cyber security, but the use and extent of surveillance technology



By Alex Scroxton, Security Editor

Published: 11 Mar 2021 11:30

A cyber attack on video surveillance startup [Verkada](#) that has seen around 150,000 video cameras, many of them in secure locations, compromised by [a hacktivist collective](#), is prompting warnings both around basic security hygiene, and [the ethics of surveillance technology](#).



Internet of Things Teddy Bear Leaked 2 Million Parent and Kids Message Recordings



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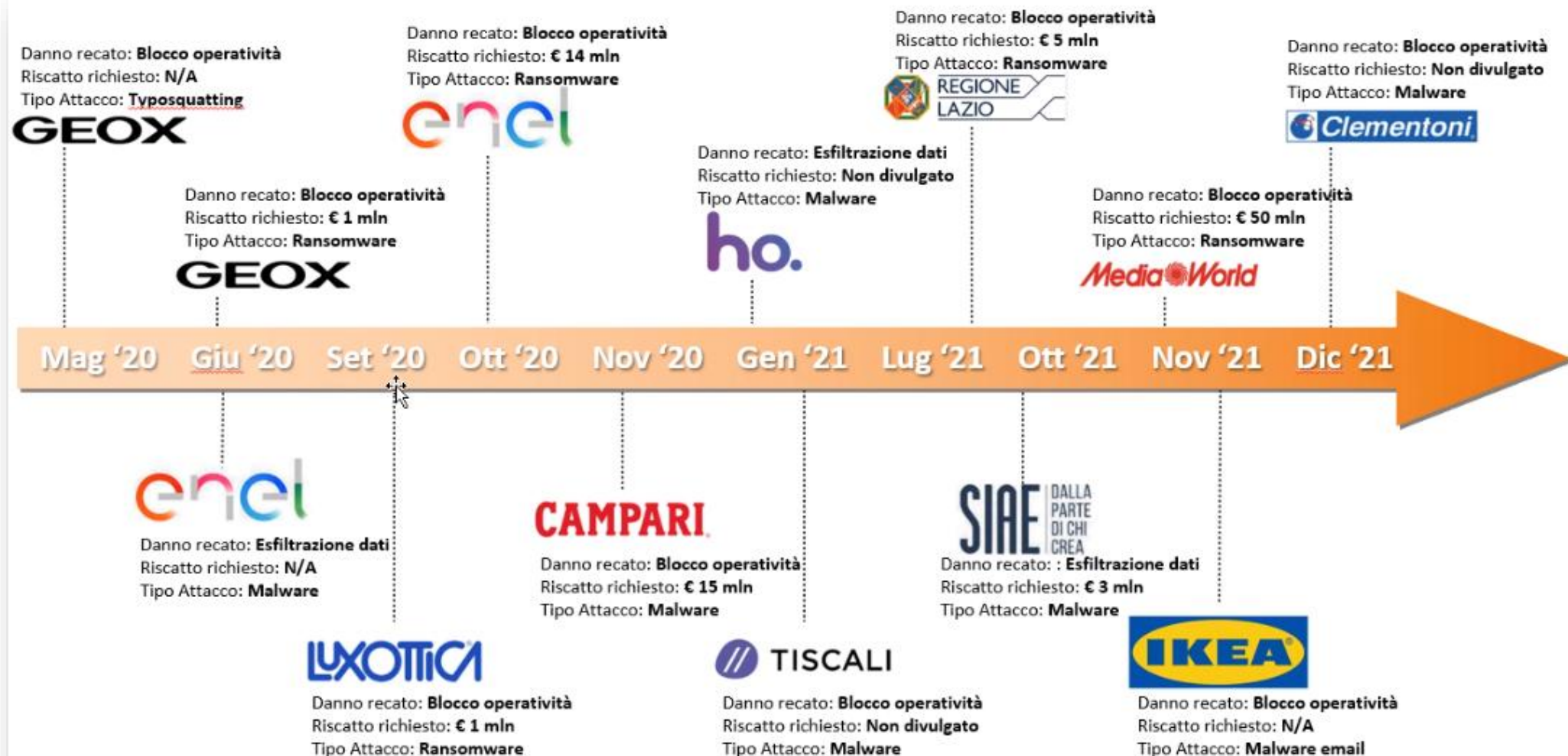
Everything is under attack... always ...

- *Cyber Attack Map*: real time maps:
 - <https://threatmap.checkpoint.com>
 - <https://cybermap.kaspersky.com>
 - ...
- The most popular ones are summarized here:
 - <https://norse-corp.com/map/>

Attack maps are visualization tools, not measurements of risk or impact.



Some significant attacks





Recent trends

- +27% severe incidents globally in 2024 vs 2023
- Italy accounts ~10% of global attacks (Rapporto CLUSIT 2025)

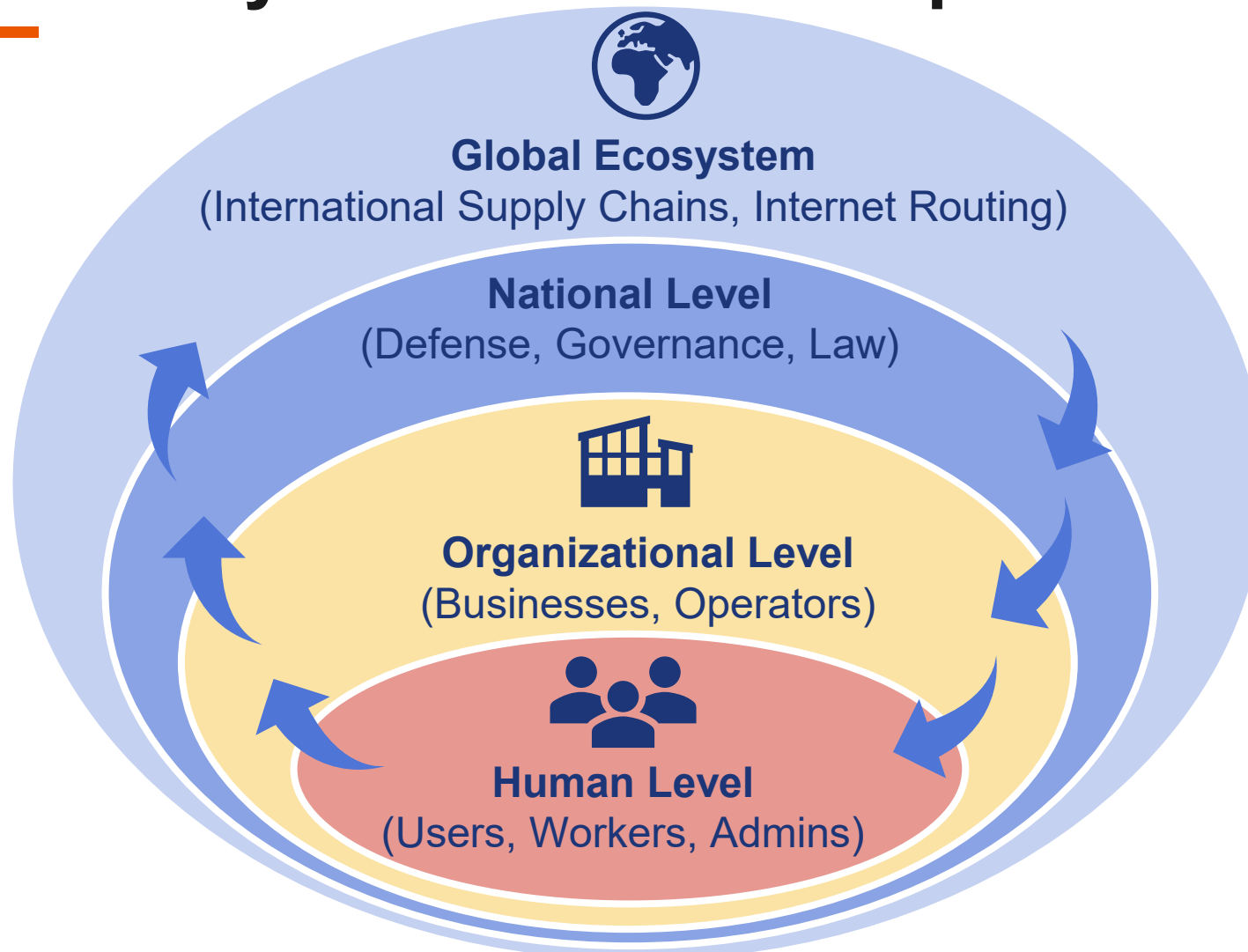
[https://clusit.it/wp-content/uploads/area_stampa/2025/Anteprima_Rapporto_Clusit_2025.pdf]



Who is affected

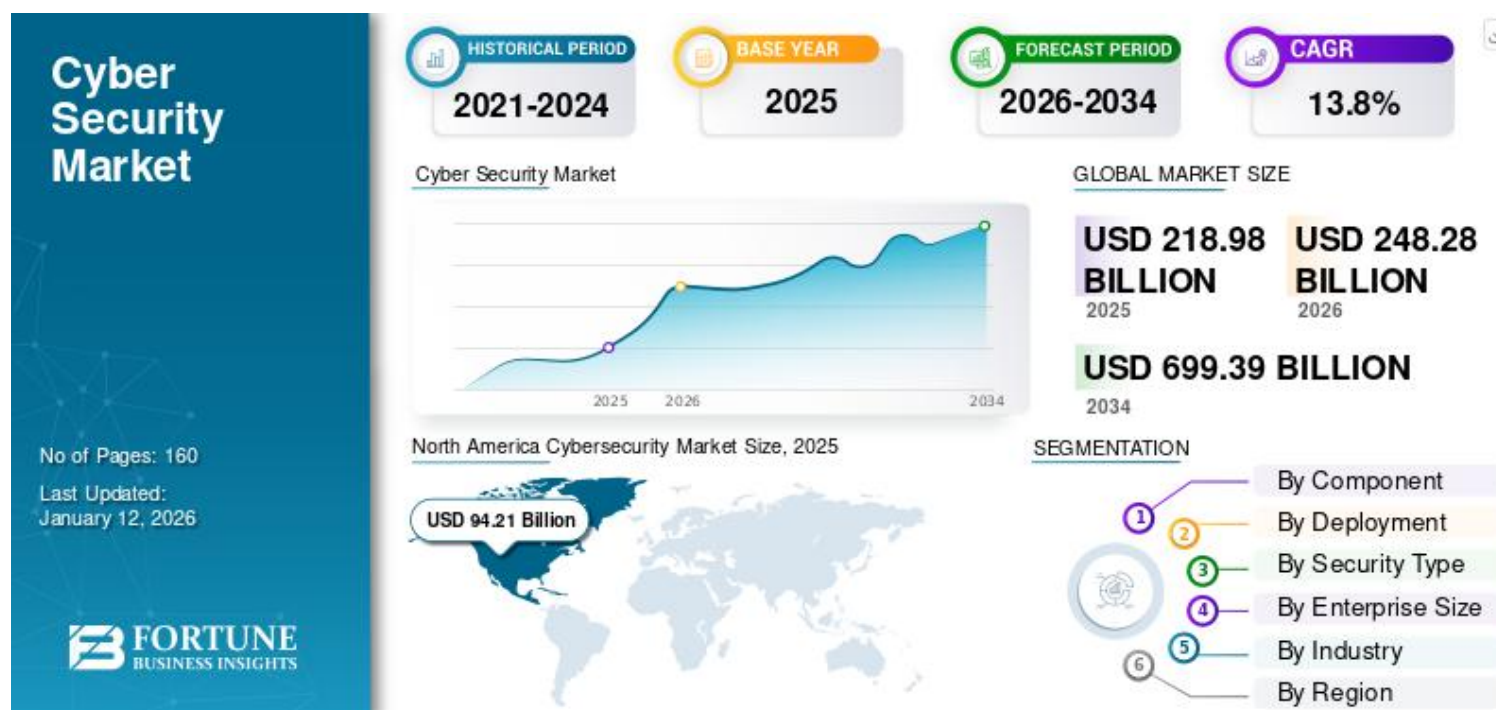
- Cybersecurity problems scale with interdependence and governance scope
- They need to be addressed at ALL levels:
 - *Individual*
 - *Organization*
 - *National*
 - *International*
- The higher the interdependence, the higher the systemic impact

Cybersecurity is a coordination problem across layers



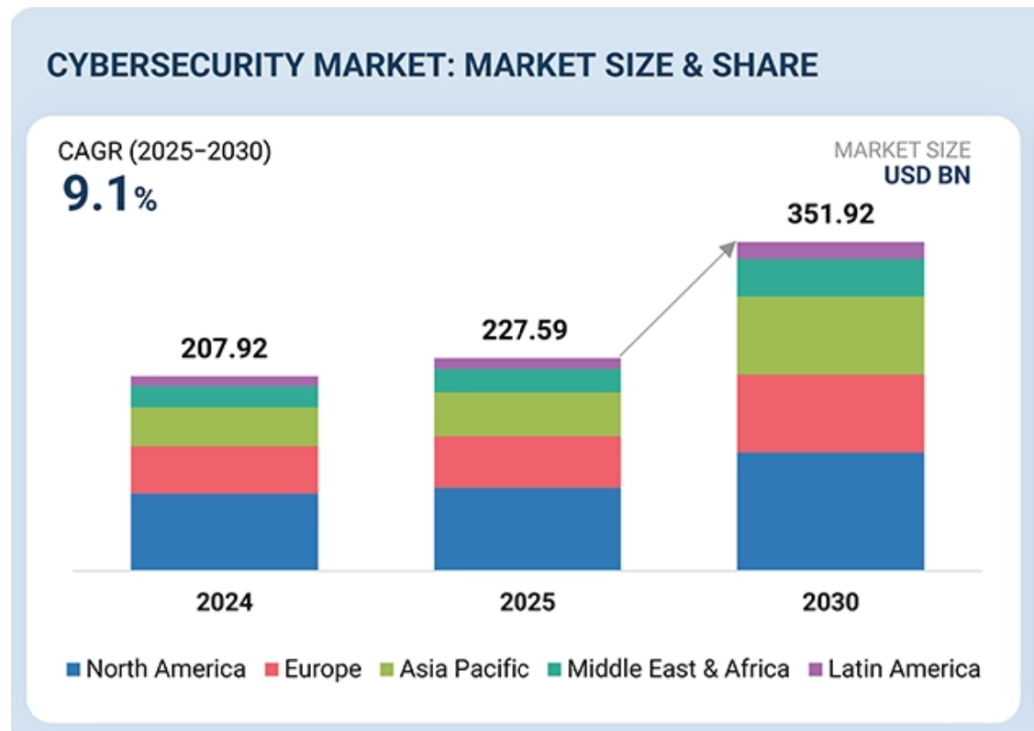
Cybersecurity – Market Global Trends

- Market value 2026: **248 Billion**
- Forecast in 2034: **699 Billion**
- <https://www.fortunebusinessinsights.com/industry-reports/cyber-security-market-101165>



Cybersecurity – Market Global Trends

- Market value 2025: **227 Billion**
- Forecast in 2030: **351 Billion**
- <https://www.marketsandmarkets.com/Market-Reports/cyber-security-market-505.html>





The limitations of the European context

- UE Market is heavily fragmented
 - Companies tend to focus on national market and have difficulties in scaling
- 67 in 500 companies with the highest growth in Cybersecurity are European
- In 2021 EU investment on cybersecurity was 814 million €
 - In the US, same year, was 15 billion €
 - Source: EIB / ECIP report (2022) – VC investments in 2021
<https://www.eib.org/attachments/lucalli/20220206-european-cybersecurity-investment-platform-en.pdf>
- This fragmentation and underinvestment translate into strategic dependency, and reduced technological sovereignty

<https://ecs-org.eu/activities/european-cybersecurity-investment-platform/>